

STUDENT NAME	TUPAKULA SRAVANTHI
SECTION	B.Sc(CS)
HALLTICKET NUMBER	0325046021

Data Analytics with Tableau

Assignment

Sheet-01

Area chart

Step-1: from the data pane, drag Order Date to the columns

Step-2: Right-click year (order date) on the column shelf

Step-3: select Month

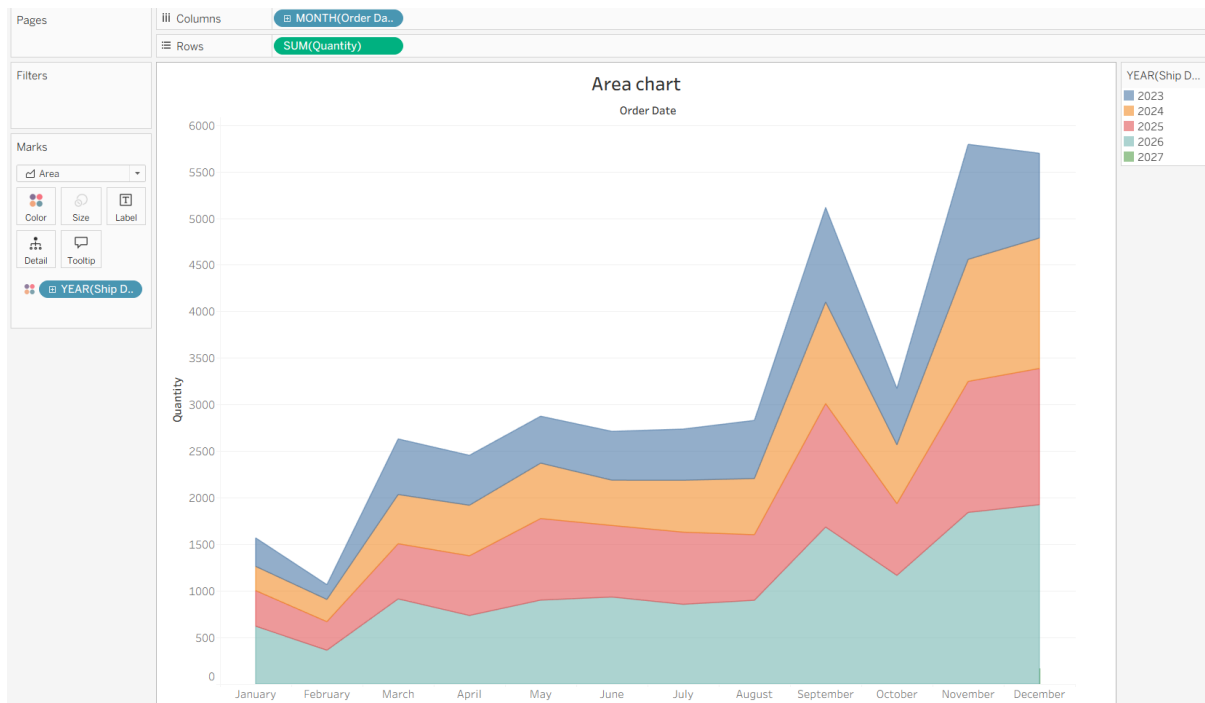
Step-4: Drag Quantity to the Rows shelf.

Step-5: Drag Ship mode to colour on the marks card

Step-6: On the marks card, open the drop-down menu

Step-7: Select Area as the chart type

Output:



Sheet-02

BAR CHART

Step-1: Drag sales to Row Shelf.

*Then tableau automatically aggregates sales as **sum (sales)**

*The tableau first shows a **Line chart** representing yearly sales trends.

Step-2: On marks card ,click the drop-down menu.

Step-3: select Bar.

Step-4: Add ship mode colours

*Drag ship mode to colour on the marks card.

Step-5: Add Region comparison

*Drag Region to the Rows shelf(left of sales).

Step-6: Filter only west Region

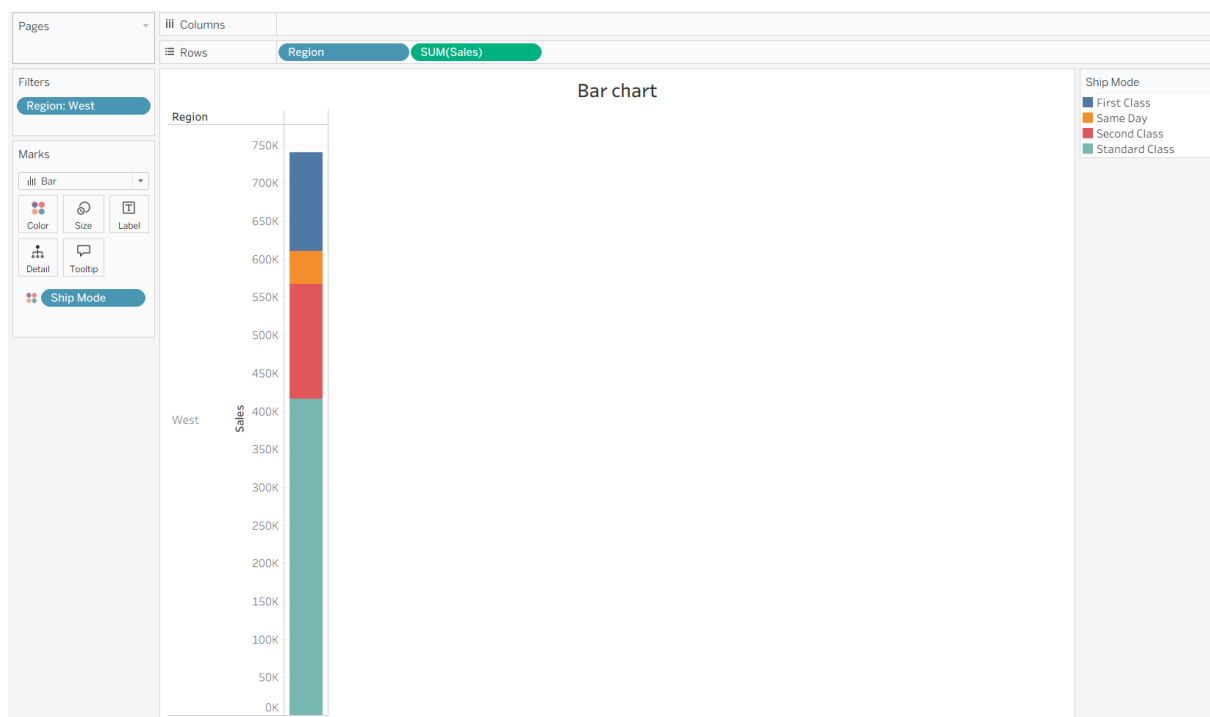
*Drag Region again to the Filter shelf.

*In Filter dialog box: uncheck **Central, East, South**

Keep only **west**

Step-7: click ok

Output:



Sheet -03

Box Plot

Step -1: Drag segment to the columns

Step-2: Drag Discount to the Rows Shelf

Step-3: click Discount on rows shelf.

Step-4: Hover over measure(sum)

Step-5: change Aggregation to Average

Step -6: Adding Region Dimension

*Drag Region to the columns shelf.

Step-7: Drop it to the right of segment.

Step-8: Click show me in toolbar

Step-9: Select Box-and-whisker plot.

Step-10: Drag Region from marks card back to columns shelf, right to segment.

Step-11: Go to Analysis menu.

Step-12: uncheck Aggregate Measures.

Step-13: click Swap rows and columns button.

Output:



Sheet-04

Gantt Chart

Step-1: Add order Date to columns

1. Drag Order Date from Dimensions.
2. Drop it on the Columns Shelf.
3. Tableau automatically shows YEAR(Order Date).

Step 2: Change Year to Week Number

1. Click drop-down arrow on YEAR(Order Date) in Columns shelf.
2. Select: More → Week Number Now columns display weekly timeline.

Step 3: Add Rows Dimensions

1. Drag Sub-Category to Rows Shelf.
2. Drag Ship Mode to Rows Shelf and place it right side of Sub-Category.

Step 4: Change Marks Type to Gantt Bar

1. In Marks card click dropdown.
2. Select Gantt Bar Now bars appear across timeline.

Step 5: Create Calculated Field We need shipping delay days.

1. Click Analysis → Create Calculated Field
2. Name: OrderUntilShip
3. Formula: DATEDIFF('day',[Order Date],[Ship Date])
4. Click OK

Step 6: Add Delay to Size

Drag OrderUntilShip to Size on Marks card. Bars become longer based on shipping days.

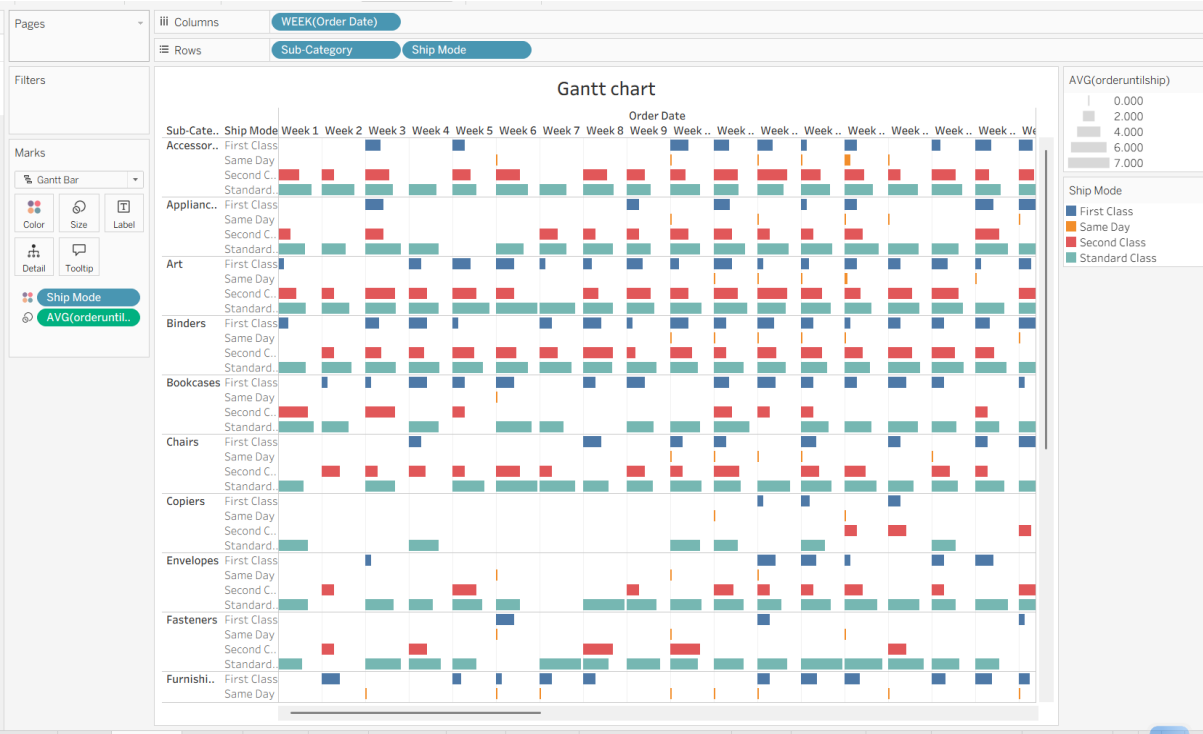
Step 7: Change SUM to AVG

1. Right-click SUM(OrderUntilShip) on Marks card
2. Select: Measure → Average
- 3.

Step-8: Add colour by ship mode

1. Drag ship mode to colour in marks card

Output:



Sheet- 05

Heat Map

Step-1: Drag segment from dimensions drop on column shelf.

Step-2: Drag Region , drop on Row shelf .

Step-3: Drag Sub-category to row shelf , place it on the right of region.

Step-4: Drag profit measures

Step-5: Drop it on colour in marks card.

Then tableau colour cells based on profits

*High profit= Blue/Dark

*loss=orange

Step-6: Click colour on marks card.

Step-7: Under Border, Choose Medium gray

Step-8: on profit color legend click dropdown arrow

*select **Edit colors**

Step-9: change the palette into Red-diverging

*check use full color range

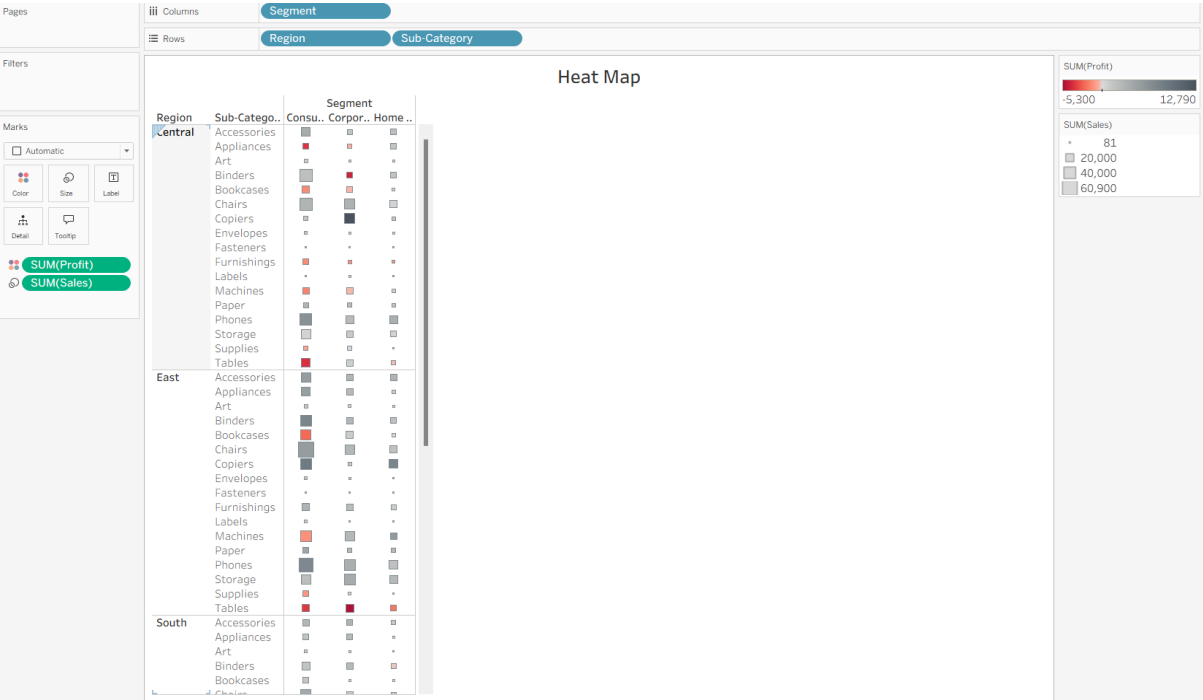
*click Apply, click Ok

Step-10: Drag sales measure, drop it on size in marks card.

Step-11: click Size on marks card

*Move slider to right until boxes to increase the box

Output:



Sheet-06

Histogram

Step-1: Drag Quantity to columns

Step-2: Select the histogram from show me.

Step-3: understand histogram

X-Axis: Quantity Bins

y-Axis: count of orders

Step-4: Drag segment and Drop on colors in marks card

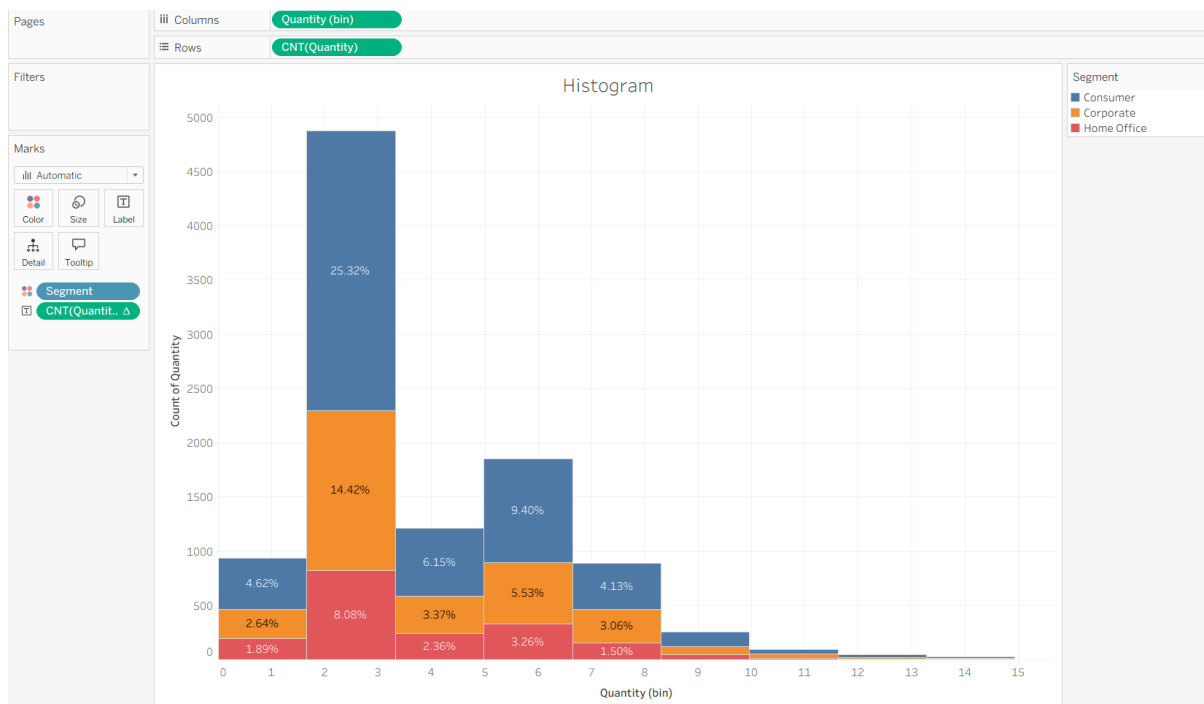
Step-5: Copy count to label

1. Hold ctrl
2. Drag CNT(Quantity) from Rows shelf to label on marks card

Step-6: Convert label to percentage of total

1. Right-click CNT(Quantity) on marks card
2. select: Quantity table calculation → percent of total

Output:



Sheet-07

Line Chart

Step-1: Drag ship date from Dimension.

*Drop it on column shelf.

Step-2: Add sales to rows

*Drag sales and drop it on Rows shelf

Step-3: Add profit to rows

*Drag profit and Drop it on Rows shelf, Right side of sales

Step-4: Blend the Axes

*Drag Sum(profit) from Rows shelf

*Drop it directly on the sales Axis

Now you will see two pale green lines

Step-5: Change year to month

*click dropdown on year(ship date) in column shelf

*select :month(from lower continuous section)

Step-6: Observe Trend

You may notice:

☒ Higher values near year end

☒ Repeating yearly spikes

☒ Seasonal business growth

Step-7: Add Forecast

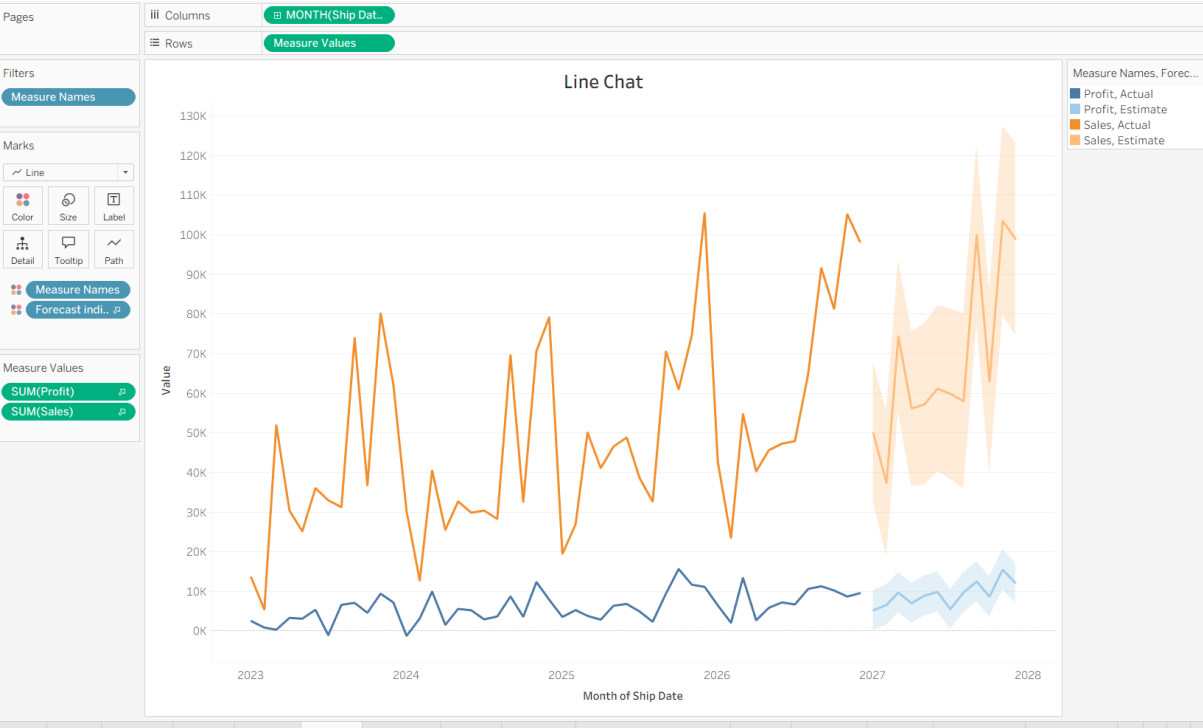
1.click Analytics pane

2.Drag Forecast

3.Drop into chart area

Tableau can automatically predicts future trends

Output:



Sheet -08

Packed Bubble Chart

Step-1: Drag category , Drop it on columns shelf

Step-2: Drag sales , Drop it on Rows shelf

*tableau creates a Bar chart by default

Step-3: Change to packed Bubbles

Step-4: click show me ,select packed bubbles

Step-5: Drag Region and drop it on details in marks card

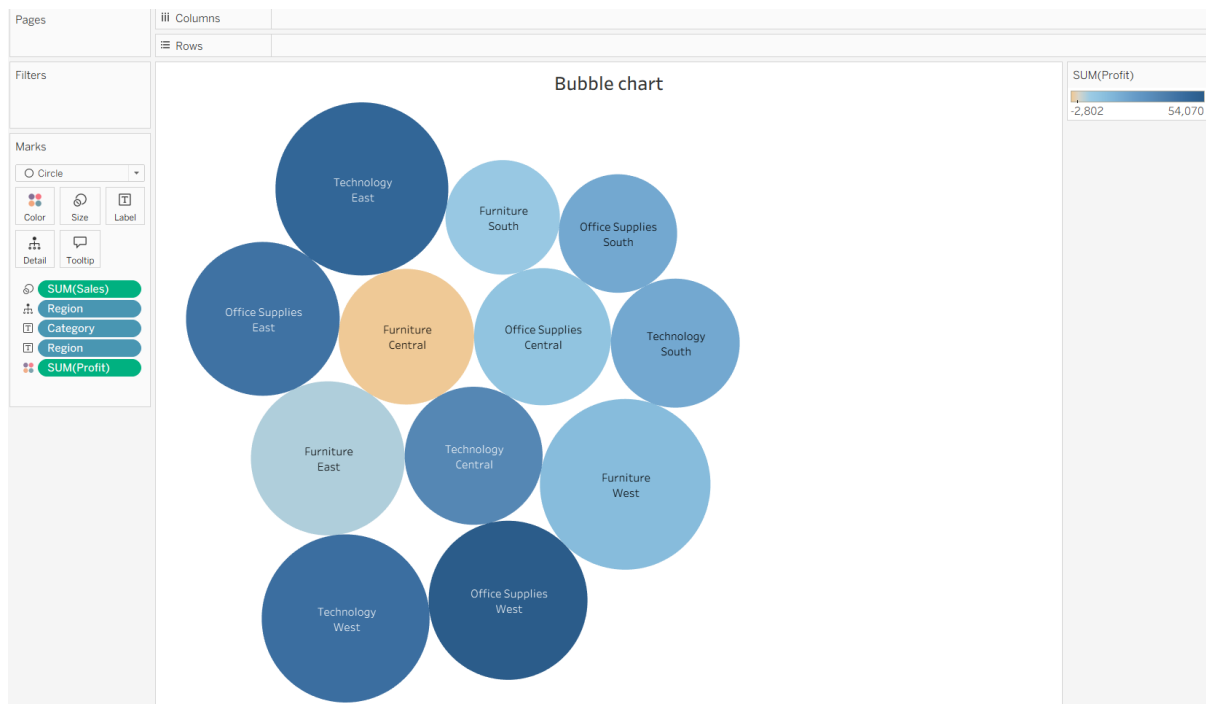
Step-6: Add profit to color

*Drag profit,Drop it on color

Step-7: Drag Region ,Drop it on Label

Now Each bubble display region name

Output:



Sheet-09

Pie Chart

Step-1: Add profit to columns

*Drag profit measure, drop it on columns Shelf

Step-2: Add category to rows

*Drag category ,drop it on rows shelf

Step-3: click show me ,select pie chart

After selecting pie chart:

☒ Mark type changes to pie

☒ Sum(profit) moves to:

Angles

Size

☒ category moves to colour

Step-4: Increase pie Chart Size

Option 1:

1.click fit menu(toolbar)

2.Select entire view

Option 2:

1.click size on Marks card

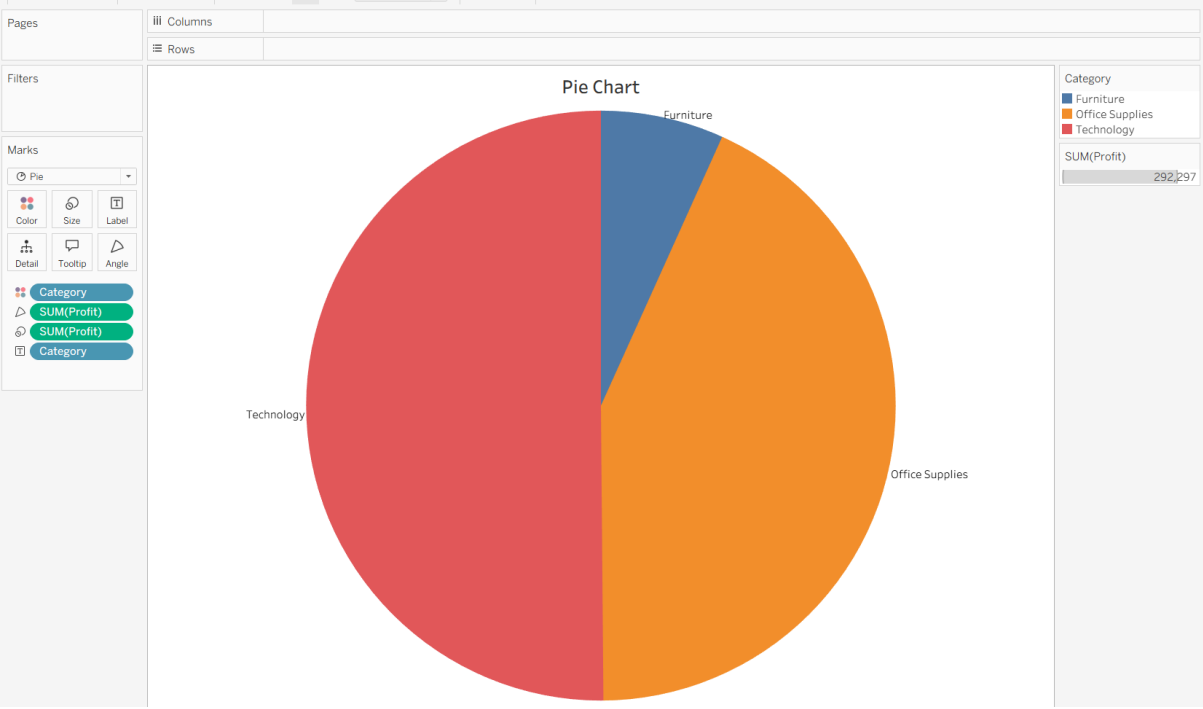
2.Move silder right

Then pie chart become larger

Step-5: Add labels

1.Drag category, Drop it on label in marks card

Output:



Sheet-10

Scatter Plot

Step-1: Add sales to columns

*Drag sales and drop it on columns shelf

Step-2: Add profits to rows

*Drag profit and drop it on Row shelf

Step-3: Add category to color

*Drag category ,drop it on color in mark card

Step-4: Add region to details

*Drag region ,drop it on details in marks card

Step-5: Adding trend lines

1.click Analytics pane

2.Drag trend line

3.Drop it on chart

Step-6: select : linear

Output:



Sheet-11

Text Table with Percentage

Step-1: Add order date to columns

*Drag order date , drop it on columns shelf

Step-2: Add sub-category to rows

*Drag Sub-category ,drop it on row shelf

Step-3: Add sales to text

*Drag sales ,drop it on text in marks card

Step- 4: Add region to rows

*Drag region , Drop it on left side of sub-category in row shelf

Step-5: convert sales to percentage

*Right-click Sum(sales)on marks card

Step-6: select Add table calculation

Step-7: Configure table calculation

In dialog Box:

Calculation type: percentage of total

Compute using: pane(down)

Step-8: then click OK

Output:

Pages

Filters

Marks

Columns

Rows

YEAR(Order Date)

Region

Sub-Category

Region	Sub-Catego...	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
	Storage	10.68%	8.10%	8.63%	9.28%
	Supplies	0.42%	0.32%	2.89%	2.98%
East	Tables	7.50%	6.67%	9.37%	7.16%
	Accessories	4.47%	11.40%	3.41%	6.85%
	Appliances	4.53%	4.26%	5.22%	5.73%
	Art	1.01%	1.13%	1.11%	1.27%
	Binders	5.40%	9.18%	10.81%	6.84%
	Bookcases	8.02%	12.51%	3.30%	3.42%
	Chairs	16.89%	12.75%	12.70%	14.50%
	Copiers	3.91%	3.16%	10.81%	11.27%
	Envelopes	0.99%	0.56%	0.71%	0.39%
	Fasteners	0.16%	0.14%	0.21%	0.13%
	Furnishings	3.32%	3.46%	5.43%	5.07%
	Labels	0.38%	0.33%	0.47%	0.42%
	Machines	10.81%	9.18%	13.80%	5.46%
	Paper	3.25%	2.53%	2.71%	3.62%
	Phones	16.04%	14.50%	11.76%	16.79%
	Storage	12.07%	8.48%	12.11%	9.49%
South	Supplies	0.36%	0.76%	1.14%	3.25%
	Tables	8.36%	5.67%	4.28%	5.48%
	Accessories	5.39%	5.80%	10.02%	6.64%
	Appliances	2.04%	5.40%	5.99%	6.47%
	Art	0.55%	1.91%	1.54%	1.05%

Text Table with Percentage

Sheet-12

Tree Map

Step-1: Add sub-category to columns

*Drag sub- category ,drop it on column shelf

Step-2: Add sales to rows

*Drag sales ,drop it on Row shelf

Step-3: Change to tree map

*Click on show me ,select tree map

Step-4: Default tree map meaning

Feature represent

Rectangle size sales

Rectangle color sales

Step-5: Add ship mode to color

*Drag ship mode, drop it on marks card

Step-6: Alternative view

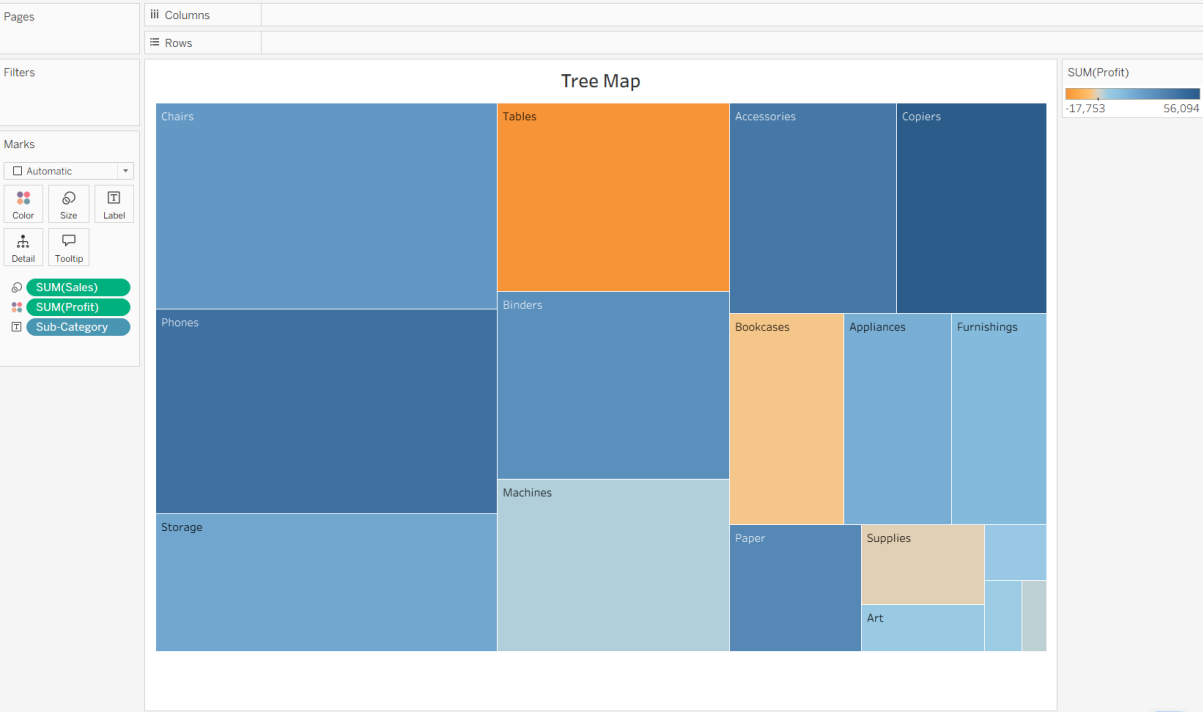
If you want profit analysis

*click undo to remove ship mode

*Drag profit

*Drop it on color

Output:



Sheet-13

Donut Chart

Step-1: Create a calculated field

- *Go to the analysis menu

- *Select create calculated field

Step-2: Name the field as zero

- *in the formula area , type: 0

- * click OK

Step-3: Drag Zero to rows

- *in measures section, drag zero to the rows shelf

Step-4: Convert into pie chart

Step-5: In marks card ,click automatic

- *select pie

Step-5: From the toolbar , choose fit →Entire view

Step-6: increase the pie size

- *in marks card ,click size

- *increase the size slider

Step-7: create second pie chart for donut hole

- *Again drag Zero from measures to the rows shelf

- *place it on the right side of first sum(zero)

Now two pie charts appear

Step-8: create dual axis

- *Right-click on second sum(zero) in rows

- *select dual axis

Now both pie charts are overlap

Step-9: Create the inner circle

- *in the marks card, click the second Sum(zero)

- *click size and reduce the size

*click color ,select white

Step-10: click the first sum(zero) in marks card

*Drag region to color

Step-11: Drag region to label on marks card

*Drag sales to label

*drag sales to angle

Step-12: click the second sum(zero)mark card

*Drag sales to label

*click label to text

Step-13: click the tree dots(...)beside text editing

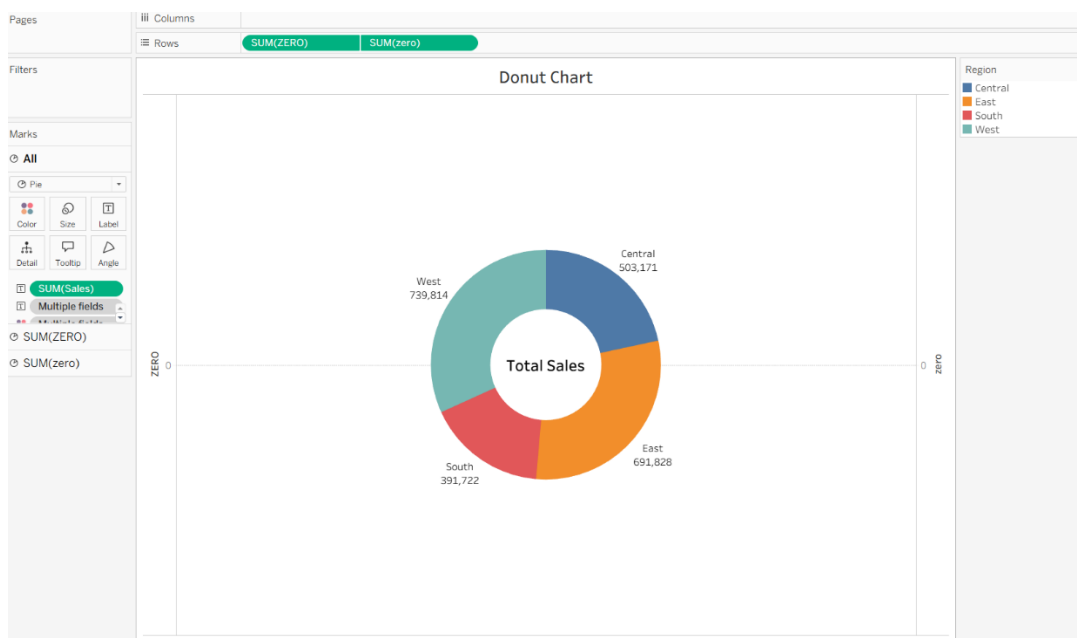
Type :

Total sales

Step-14: click OK

Now the center of the donut chart display total sales

Output:



Sheet-14:

Map chart

Step-1: create the basic Map

- *From data pane ,drag state to the worksheet

- *tableau automatically generate a map using latitude and longitude

Step-2: Drag sales from measure to color in the marks card

- *the map now display sales value with diff color

Step-3: in marks card, click automatic

- *select edit color

- *choose a color palette(purple)

- *click OK

Step-4: Drag state to label, drag sales to label

Now each state show its values

Step-5: Drag region to filter shelf

- *select all region

- *click OK

Step-6: Drag profit to tooltip

Over any state to view sales and profit details

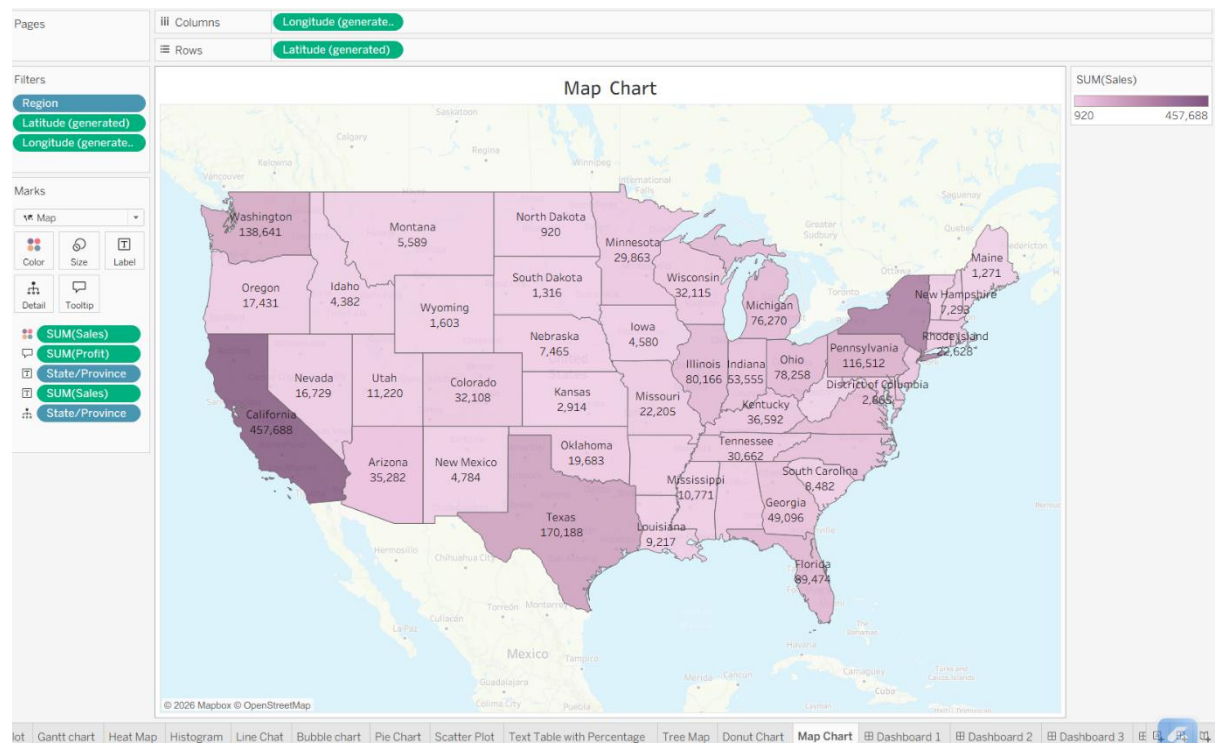
*Step-8: go to map →map layer

- *Adjust :

- Background style (out doors)

- Washout(70%)

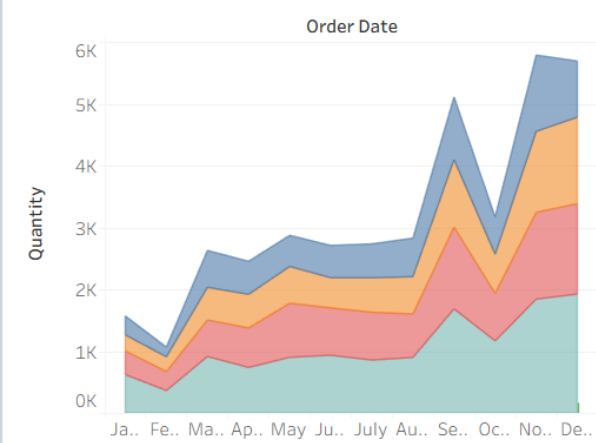
Output:



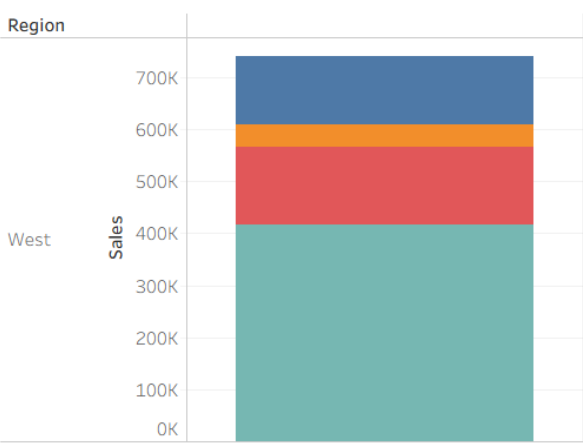
DashBoard-1

lew Data Source (Ctrl+D)

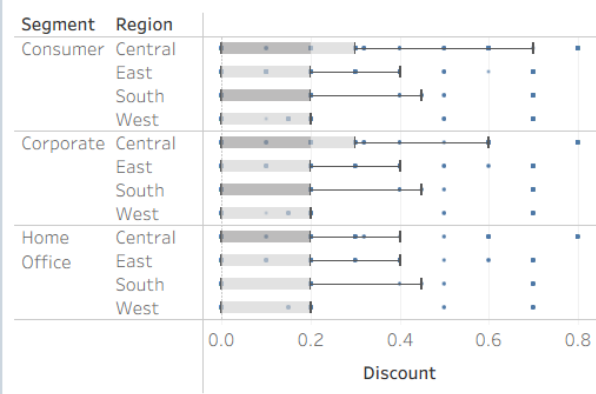
Area chart



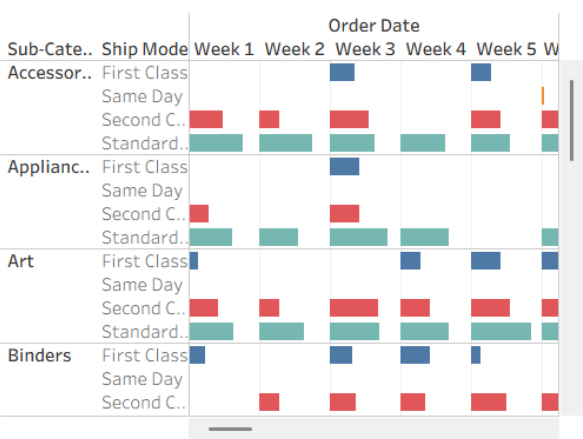
Bar chart



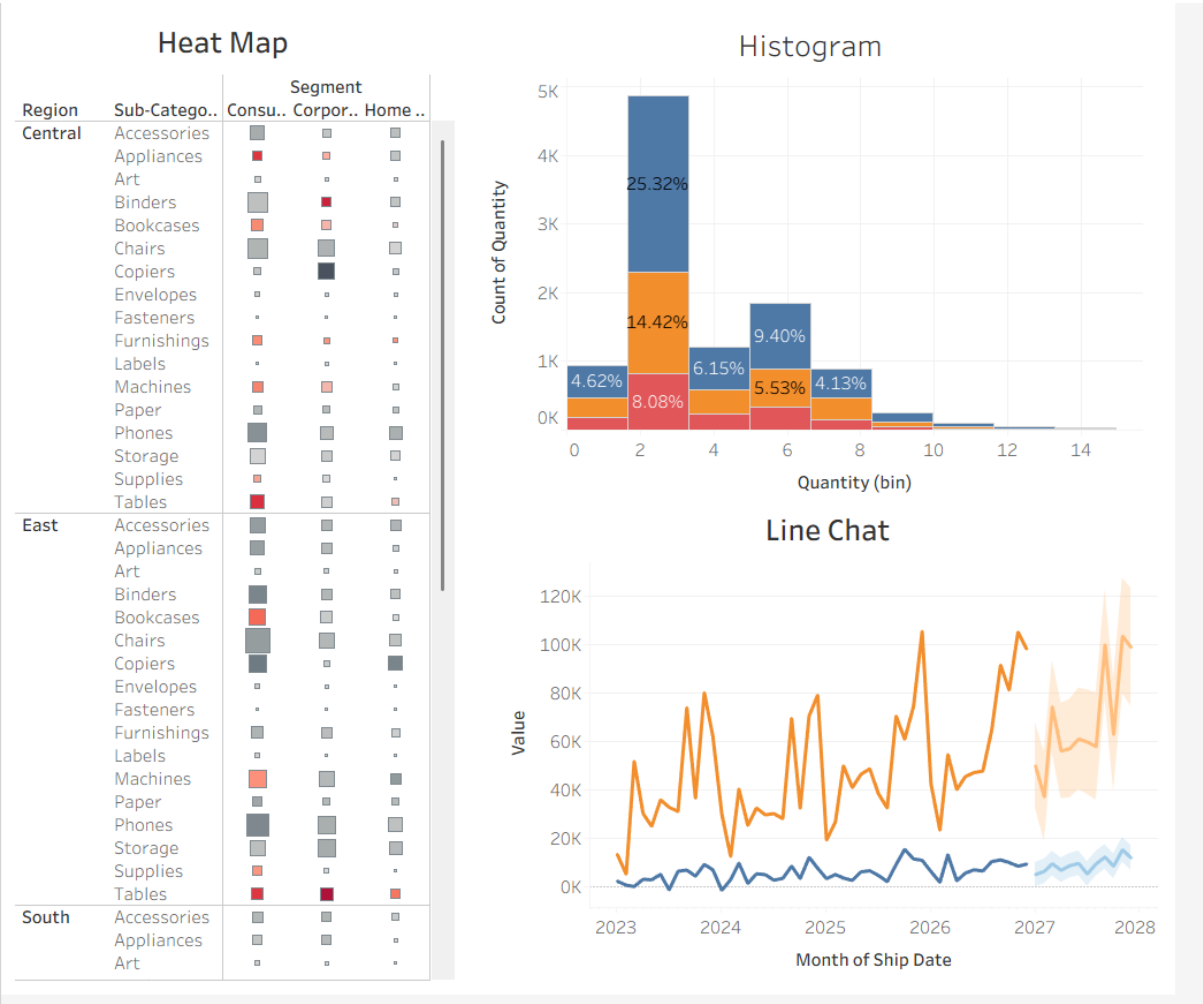
Box plot



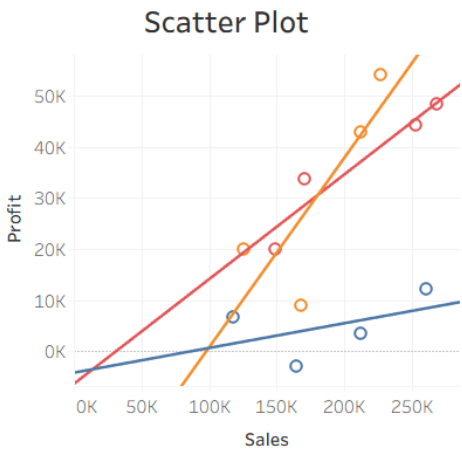
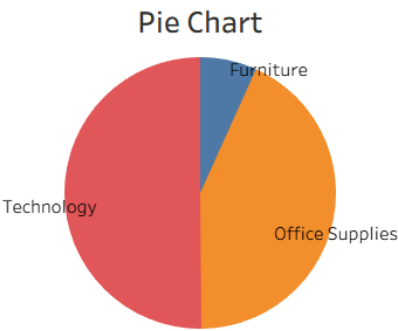
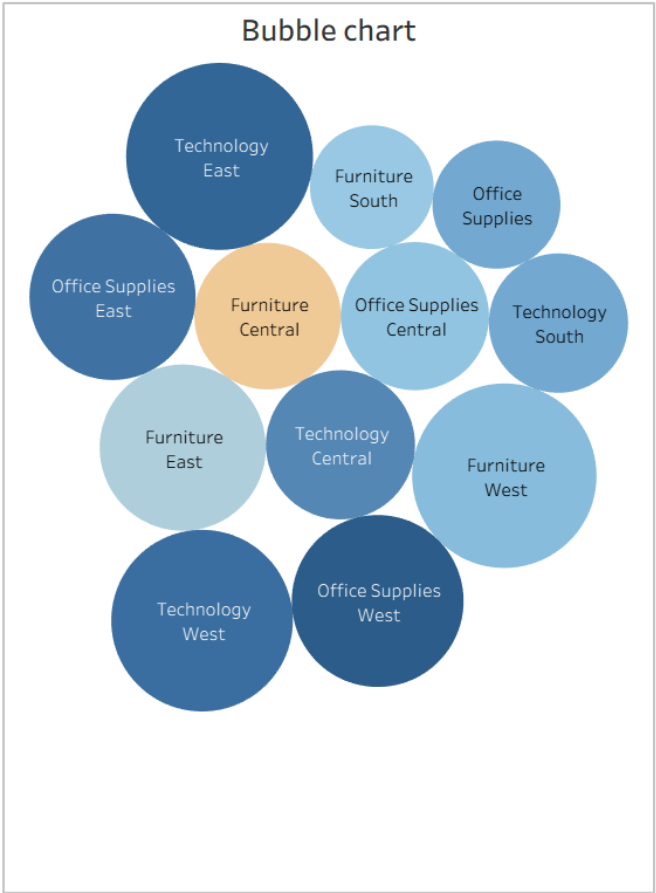
Gantt chart



DashBoard-2



DashBoard-3

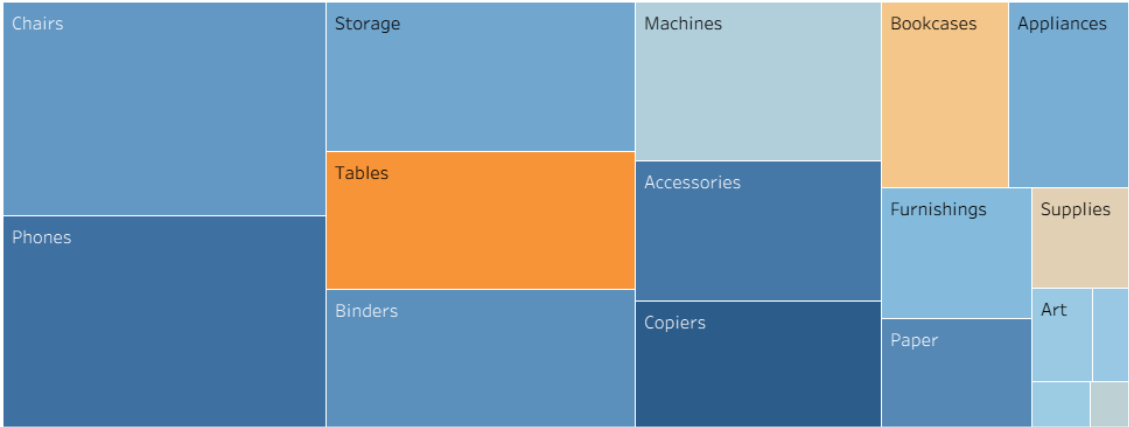


DashBoard-4

Text Table with Percentage

Region	Sub-Catego..	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
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	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.25%	2.45%	2.61%	4.22%

Tree Map



DashBoard-5

